



SIMATS ENGINEERING

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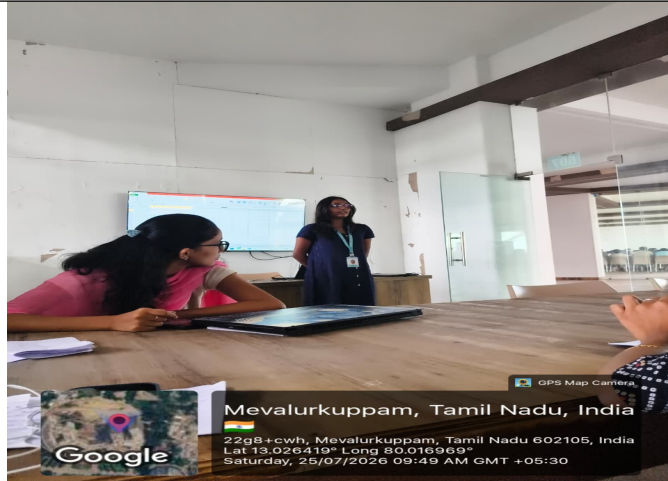
Course Code: ITA1404

Slot: C

Course Name: Ethical Hacking

Course Faculty: DR RameshKumar K

Project Title : Ethical Hacking



Project Description: here you write what you did in this project including Model Description

The project titled "**Ethical Hacking: Securing Digital Systems Through Authorized Security Testing**" focuses on understanding the principles, techniques, and importance of ethical hacking in modern cybersecurity. The project explains how ethical hackers identify vulnerabilities in computer systems, networks, and applications with proper authorization to prevent cyberattacks and protect sensitive information. It highlights the role of ethical hackers, also known as white-hat hackers, in strengthening an organization's security by performing vulnerability assessments, penetration testing, and security analysis.

The project demonstrates the ethical hacking process, including reconnaissance, scanning, vulnerability identification, controlled exploitation, reporting, and recommending security improvements. It also emphasizes the ethical and legal responsibilities of security professionals, ensuring that all testing is performed with permission while maintaining confidentiality and integrity. The project discusses common security threats such as malware, phishing, ransomware, and unauthorized access, along with the preventive measures used to defend against them.